Seat No _____

22 FEB 2022

BCA Semester – 3 Examination

Paper Name: Operating System | Paper Code: (22691)

Time duration: 2 Hours | Total Marks: 56

Que - 1 Define Operating system. List out and explain different types of operating system. [14]

OR

Que – 1

(A) Define functions of operating system. [7]

(B) Explain monolithic structure of operating system. [7]

Que – 2 List out and explain different process scheduling algorithm with proper example [14]

OR

Que – 2

(A) Define schedular. List out and explain different types of schedulers in detail. [7]

(B) Define following terms: process, cpu utilization, throughput, turn around time, waiting time, response time. [7]

Que – 3 Define deadlock. Explain Deadlock avoidance using banker's algorithm with proper example. [14]

OR

Que – 3

(A) Explain deadlock prevention techniques in detail. [7]

(B) Write a note on deadlock detection and recovery [7]

Que – 4 Define page fault and page replacement? For the following page strings find out number of page faults with the FIFO, LRU and OPTICAL page replacement algorithm and conclude which algorithm is best suitable. [14]

Page string: 7,0,1,2,0,3,0,4,2,3,0,3,2,1,2,0,1,7,0,1 of frame size 4

OR

Que – 4

(A) Explain paging with proper example. [7]

(B) Explain continuous memory allocation technique. [7]